

## GROUP 4

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**PROJECT TITLE:** NimbusDesk - *Technical Support and Help Desk*



### PROJECT OVERVIEW / DESCRIPTION:

NimbusDesk is a web-based technical support and help desk system designed specifically to streamline IT incident management within aviation and airline operations. The system addresses the critical problem of delayed technical support and lost maintenance requests caused by manual or unorganized reporting methods. Aviation staff can use the portal to submit support tickets for software glitches, network failures, or broken hardware, tracking the resolution progress in real time. It is primarily intended for airline IT support staff, system administrators, and aviation employees who need immediate technical assistance. By centralizing all technical reports into a single organized dashboard, NimbusDesk improves response time, ensures proper incident documentation, and minimizes operational downtime in time in fast-paced airport environments.

## **PROBLEM STATEMENT:**

1. Aviation employees currently lack a centralized digital platform to report IT incidents, leading to technical problems being overlooked or improperly documented from the time they are reported until resolution.
2. IT support personnel struggle to prioritize and monitor active technical issues due to inefficient manual logging, resulting in delayed response times that can disrupt airline or airport operations.
3. There is no reliable way to track the historical frequency of specific hardware or network failures, making it difficult for administrators to identify recurring system vulnerabilities.

## **OBJECTIVES:**

### **General Objective:**

To develop a web-based technical support and help desk ticketing system that centralizes and tracks IT incident reporting for aviation operations.

### **Specific Objectives**

1. To provide a digital ticketing module where aviation employees can easily report software glitches, network issues, and hardware failures.
2. To allow IT support personnel to view, prioritize, and update the resolution status of incoming technical requests in real-time.
3. To generate basic summary logs of reported incidents to help administrators monitor common technical problems and IT response activity.

## Target Users

- Airline / Airport Employees (End-users)
- IT Help Desk Technicians
- IT Supervisors / System Administrators

The primary users of NimbusDesk are general aviation employees who will use the portal to submit technical support tickets, and IT help desk technicians who will use the system to manage, update, and resolve those reported incidents.

## Proposed Features

| No. | Feature                                              | Description                                                                                                                                        |
|-----|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Role-Based Access Control (RBAC) User Authentication | Provides a login system with distinct access levels, separating standard aviation employees (ticket creators) from IT personnel (ticket managers). |
| 2   | Ticket Submission Portal                             | Allows aviation staff to create a new support request by inputting the issue category, a brief description, and the urgency level.                 |
| 3   | Help Desk Dashboard                                  | Displays a quick overview of active, pending, and resolved support tickets relevant to the user's role.                                            |
| 4   | Ticket Status Management                             | Enables IT admins to assign tickets to technicians, update the progress status (e.g., Open, In Progress, Resolved), and add resolution notes.      |
| 5   | Incident History Log                                 | Maintains a basic, searchable record of all past resolved tickets and technical reports for future reference and simple monitoring.                |